Math 271 Mathematical Methods in Economics, Fall 2026

**Tutorial - 4**

*Topic: Revision for Mid-Term II*

In this tutorial worksheet, we will try to prove the following two theorems to practice for the upcoming mid-term.

### Question 1. On Uniform Continuity and Cauchy Sequences

Recall from Homework 3, the definition of a Cauchy Sequence

> **Cauchy Sequence**: A sequence of real numbers $\{x_n\}$ is a Cauchy sequence if for any $\varepsilon > 0$ there is an integer $N$ such that for all $i, j \geq N; |x_i - x_j| < \varepsilon$

Also, recall from Homework 4, the definition of Uniform Continuity:

> **Uniform Continuity**: We say that $f : A \to \mathbb{R}$ is uniformly continuous on $A$ if given any $\varepsilon > 0$ there is a $\delta(\varepsilon) > 0$ such that if $x, y \in A$ are any reals satisfying $|x - y| < \delta(\varepsilon)$, then $|f(x) - f(y)| < \varepsilon$

The two notions are interconnected as follows:

**Theorem**: If $f : A \to \mathbb{R}$ is uniformly continuous on $A \subset \mathbb{R}$ and if $\{z_n\}$ is any Cauchy sequence in $A$, then, $\{f(x_n)\}$ is a Cauchy sequence in $\mathbb{R}$.

*Prove the above theorem.*

### Question 2. On Characterization of Differentiable Quasi-Concave functions

Over the years, mathematical economists have provided multiple characterizations for quasi-concave functions. The one given below[1] uses the following notion:

> **Semistrict Local Maximum**: Let $\xi : C \to \mathbb{R}$ be a real valued function defined on the open interval $C \subset \mathbb{R}$. Then $\xi$ is said to attain a *semi-strict local minimum* at a point $p^0 \in C$ if there exist two points $p^1 \in C$, $p^2 \in C$, $p^1 < p^0 < p^2$, such that $\xi(p^0) \leq \xi(\lambda p^1 + (1 - \lambda)p^2)$ for every $0 \leq \lambda \leq 1$ and $\xi(p^0) < \min[\xi(p^1), \xi(p^2)]$.

[1] DIEWERT, W. E., AVRIEL, M., and ZANG, 1. (1981), Nine kinds of quasiconcavity and concavity, J. Econ. Theory 25, 397-420

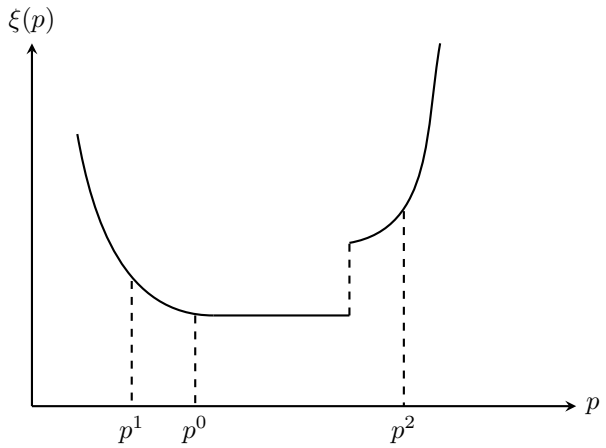
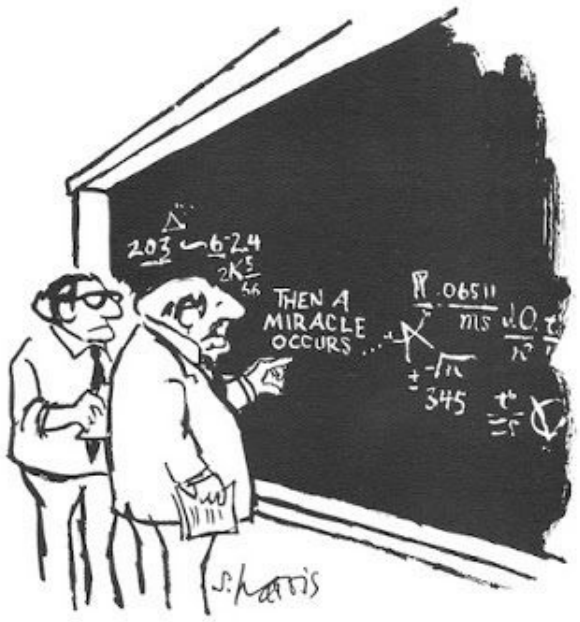



Figure 1: Semistrict local minimum.

**Theorem (Diewert, Avriel, and Zang, 1981)**: Let $f : X \to \mathbb{R}$ be a differentiable function defined on the open convex set $X \subset \mathbb{R}^n$. Then $f$ is quasi-concave if and only if for every $x^0 \in X$ and $v \in \mathbb{R}^n$ such that $v^\top v = 1$ and $v^\top \nabla f(x^0) = 0$, the real valued function $F : \mathbb{R} \to \mathbb{R}$, defined by $F(t) = f(x^0 + tv)$ does not attain a *semistrict local minimum* at $t = 0$

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"I THINK YOU SHOULD BE MORE EXPLICIT HERE IN STEP TWO."